

RAMZI ABIDI

ROBOTIC AND AUTOMATION ENGINEER

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As a passionate Robotic and automation Engineer naturally driven toward research & innovation, dynamic and motivated by new technologies, passionate about robotics, programming, embedded systems, mechanical design, and industrial computing.

PROFESSIONAL EXPERIENCES

PLC Programmer Engineer, Mondragon Assembly

Jun 2022 to Present

Tunisia

Main activities : ✓ Program Beckhoff PLC . ✓ Setup Machines and Commissioning . ✓ Works with HMI , Inverters , Beckhoff PLC , Robot , Cameras .

During my time at the Mondragon Assembly, I have been involved in many of the following missions:

✓ Installation and commissioning phase of BVG project - India

- Make dry mode to test the machine .
- Commissioning with Materials
- Apply the modification of PID for MTS
- Configuration and commissioning of 6 Axis Robot “ESTUN Robot “ and program change and adapt communication between MTS and Layup .
- Camera calibration and work with vision for Estun Robot .

✓ Setup of MTS 3000 for Reden Project - Spain

- Configuration different stations of machines (Scara Robot , Camera , Axis , Drive, inverters ..) .
- Make commissioning of machine

✓ Participate in Setup of six machines MTS 3000 for BVG project - Spain

- Configuration different stations of machines (Scara Robot , Camera , Axis , Drive, inverters ..) .
- Make commissioning and validation of machines .

✓ Support of client Zerrgoun green energy – Algeria

- Assistance with client to ramp up production .
- Training for the control of various machines.
- Make an upgrade of the line .
- Setup and teaching of 6 Axis ABB Robot .

✓ Modification on the line of client MTA- Morocco

- Add robotic Arm to the existing machine as a result to allow making tests for electronic cards .
- Make the HMI modifications.

Electrical and Automation Manager, Poulina Group Holding , Carthago Ceramic Jbel oust

Jan 2021 to May 2022

Zaghouan

Managed the electrical and automation maintenance team, ensuring efficient operations and high-quality output.

- Developed and implemented preventive and corrective maintenance plans to enhance production reliability.
- Supervised the integration and training of new recruits, ensuring they were well-versed in machine operations.
- Collaborated with subcontractors and suppliers to validate spare parts and oversee machine modifications.

Project Engineer, Poulina Group Holding , Carthago Ceramic Jbel oust**Feb 2020 to Jan 2021**

Zaghuan

Managed project resources effectively, optimizing technical, human, and financial inputs.

- Planned and scheduled assembly work, ensuring strict adherence to deadlines.
- Oversaw modifications during assembly, collaborating closely with Engineers and technicians from SACMI, SYSTEM, and LB.

EDUCATION

National Higher School of Engineers of Tunis**2023 to Jan 2025**

Master degree

Speciality : Advanced robotic systems Engineering and Artificial Intelligence

National School of Engineers of Gafsa**2016 to Jan 2019**

National diploma of Engineering

equivalent to Master Degree Speciality : Electromechanical engineering

Higher institute of Technological Studies of RADES**2013 to Jan 2016**

Bachelor degree in electrical engineering

Speciality : Automation and industrial IT

Technical High School 9 April 1938, Jendouba**2012 to Jan 2013**

Baccalaureate

Speciality : Technical Sciences

INTERNSHIPS

Master Thesis, AVIONAV**Mar 2025 to Dec 2025**

Tunis

Integration of an Automated Defence module on the NOMAD Rover : "Design of the SAMURAI System"

- Rover Model Integration in ROS 2 and development of the robotic architecture
- Rover Navigation and Control using ROS 2
- Multi-Sensor Integration: LiDAR, RGB-D Camera, Infrared (IR) Camera, and IMU
- Computer Vision & AI for automated threat detection and recognition using a military dataset.
- Environment Perception and Mapping using onboard sensors
- Simulation, Testing, and Validation of the integrated system

End of study internship, TIRA ROBOTS**Feb 2019 to Jun 2019**

Subject: Study , design et simulation of SCARA robot Tasks: ✓ Study of the existing ✓ Geometric , Kinematic and dynamic modeling. ✓ Choice of materials and robot design . ✓ Numerical simulation and project validation .

Engineer internship, Mechanical workshops of the Sahel (AMS)**Jul 2018****Engineer internship, General Northern industry GINOR****Jun 2017 to Jul 2017****End of study internship, The official printing press of the Tunisian Republic****Feb 2016 to May 2016**

Subject: Automation of the Rima System stacker machine. Tasks: ✓ Study of the existing and choice of solutions. ✓ PLC selection. ✓ Automation and supervision of the machine using a HMI with TIA Portal software .

Optional internship, Electric Chiheb**Jan 2016****Technician internship, Asteel flash electronic Tunis****Jan 2015****Internship, STEG****Jan 2014**

PROJECTS AND ACADEMIC EXPERIENCES

Study , Design and prototyping of a SCARA Robot

Establishment : National Higher School of Engineers of Tunis ✓ Geometric , Kinematic and dynamic modeling. ✓ Choice of materials and robot design . ✓ Building computer vision algorithm using AI based on Yolo V8 algorithm for object detection. ✓ Prototyping using 3D printer and validate the project (project under development) .

Study , Design and prototyping of a DELTA Robot

Establishment : National Higher School of Engineers of Tunis ✓ Geometric , Kinematic and dynamic modeling. ✓ Choice of materials and robot design . ✓ Building computer vision algorithm using AI based on Yolo V8 algorithm for object detection. ✓ Prototyping using 3D printer and validate the project (project under development) .

Study and realization of a line follower robot base on PIC 16F877

Establishment : higher institute of technological studies of RADES

- Detailed study on the microcontroller.
- Choice of used sensors (LDR).
- Electrical schematic preparation on ISIS and card programming

Study and design of a robot arm controlled remotely

Establishment : National School of Engineers of Gafsa

- Study of robotic arms in the market.
- Choice of materials and electrical design.
- Mechanical design of the arm.

PROJECTS

Realization of an all terrain robot controlled by Bluetooth

Creation of an autonomous line follower robot based on Arduino

Realization of a Sumo robot for a robotic competition.

Implementing an intelligent garbage (IOT challenge by sofecom)

- Choice of materials and study on the sensors used.
- Electrical wiring and programming.
- Prototype test .

Design and construction of a 4 quadrant chopper

- Study and Choice of materials
- Realization of electrical schematics of cards using ISIS and ARES
- Printing of electronic cards.
- Test and validation

Design and construction of a wheeled robot controlled from a LabVIEW GUI Against Corona virus

The main idea was to make the communication between the covid-19 patients and the doctors easier and avoid any physical contact with the infected patient . The following features were implemented and can be controlled from the labVIEW interface : -DC Motors with a regulated braking system -Wifi Camera -A controlled disinfection system -Also three trays for carrying food and beverage to the patient .

Design and construction of LOW cost ventilator against corona Virus

This project is the process of converting a low-cost blower into a rudimentary non-invasive pressure support ventilator that could help with breathing during respiratory distress . It has 2 mode , CPAP (Continuous Positive Airway Pressure) and BiPAP (Bilevel Positive Airway Pressure). “ This project was an implementation of an open source project”.

SKILLS

- Twincat3
- TIA PORTAL
- Beckhoff PLC
- Robot Programming

- Computer vision
- ISIS
- Matlab
- C /C++
- PIC
- Excel
- PLC
- SolidWorks
- Arduino
- Embedded C
- Raspberry
- powerpoint
- Machine learning
- ROS 2
- Catia V5
- VHDL
- STM32
- SysML
- Python
- MicroC
- Ansys
- Labview
- WORD
- Kinematics

ACTIVITIES

Member, electrical engineering club in ISET of rades

Vice President, engineering geeks club EGC

Design and prototyping of Robots

Participation in robotics competitions in Tunisia

Organization of training sessions (robotics ,electronics ...)

Animate training at Maker Skills

LICENSES & CERTIFICATIONS

EF SET English Certificate (C1 Advanced), EF SET	Mar 2022 to Mar 2022
PLC Basics Certification, Beckhoff Automation	Jan 2023 to Jan 2023
Twincat3 :Getting started , Beckhoff Automation	Jan 2023 to Jan 2023
EtherCAT , Beckhoff Automation	Jan 2023 to Jan 2023
TwinSAFE Basic, Beckhoff Automation	Jan 2023 to Jan 2023
Python 101 for Data Science , Cognitive Class	Dec 2023 to Dec 2023
Intoduction to Cybersecurity, Cisco Networking Academy	Dec 2024 to Dec 2024
Code Foundation for ROS, The Construct Robotics Institute	Feb 2025 to Feb 2025
Machine Learning with Python: Decision Trees, LinkedIn	Nov 2025 to Nov 2025
Machine Learning with Python: k-Means Clustering, LinkedIn	Nov 2025 to Nov 2025
Machine Learning with Python: Association Rules, LinkedIn	Nov 2025 to Nov 2025
Introducing Robotic Process Automation, LinkedIn	Nov 2025 to Nov 2025
Process Mining for Robotic Process Automation, LinkedIn	Nov 2025 to Nov 2025
RPA, AI, and Cognitive Tech for Leaders, LinkedIn	Nov 2025 to Nov 2025
Learning Industrial Automation, LinkedIn	Nov 2025 to Nov 2025
Learning PLC Ladder Logic: 1 The Basics, LinkedIn	Nov 2025 to Nov 2025
Learning PLC Ladder Logic: 2 Diving Deeper, LinkedIn	Nov 2025 to Nov 2025
PLC Memory Organization, LinkedIn	Nov 2025 to Nov 2025
PLC Sequencer Logic, LinkedIn	Nov 2025 to Nov 2025
PLC Program Flow and Control Instructions, LinkedIn	Nov 2025 to Nov 2025
PLC Simulation Software: Factory I/O with Connected Component Workbench, LinkedIn	Nov 2025 to Nov 2025
Machine Learning with Python: Logistic Regression, LinkedIn	Nov 2025 to Nov 2025

Machine Learning and AI Foundations: Causal Inference and Modeling, LinkedIn	Nov 2025 to Nov 2025
Machine Learning and AI Foundations: Prediction, Causation, and Statistical Inference , LinkedIn	Nov 2025 to Nov 2025
Machine Learning and AI Foundations: Producing Explainable AI (XAI) and Interpretable Machine Learning Solutions, LinkedIn	Nov 2025 to Nov 2025
Machine Learning and AI Foundations: Decision Trees with KNIME, LinkedIn	Nov 2025 to Nov 2025

VOLUNTEERING

Training for robotics enthusiast, The Gafsa University Center for Cultural and Sports Activities **Jan 2017 to Dec 2018**

Gafsa

An immersive Arduino training session , by explaining the basics of Arduino, basic electronics , motor control, and communication protocols (Bluetooth and WiFi) through hands-on coding examples. This session was done for engineering students and hobbyists eager to enhance their skills and develop innovative projects. The goal at the end is to design and implement your own Arduino-based creations.

Doctor J robot, Higher institute of Technological Studies of Jendouba **Mar 2020 to Apr 2020**

Jendouba

Design and construction of a wheeled robot controlled from a LabVIEW GUI Against Corona virus in collaboration with other Engineers

The main idea was to make the communication between the COVID-19 patients and the doctors easier and avoid any physical contact with the infected patient .

The following features were implemented and can be controlled from the labVIEW GUI :

- DC Motors with a regulated braking system
- WiFi Camera with a two way communication
- A controlled disinfection system
- Also three trays for carrying food and beverage to the patient .

HOBBIES

Gaming, Reading Books , building Robots